A Spatiotemporal, Physics-Constrained, Uncertainty-Quantified Graph Kolmogorov-Arnold Network for Multi-Station, Multi-Task Forecasting and Symbolic Discovery of Localized Geomagnetically Induced Current Risk

System Architecture Overview

The forecasting of localized geomagnetically induced currents (GICs) and geoelectric field enhancements during severe space weather events represents a critical challenge in upper atmospheric physics and power grid resilience. Traditional global magnetohydrodynamic (MHD) models provide physically consistent simulations of solar wind-magnetosphere-ionosphere (SW-M-I) coupling but are computationally prohibitive for real-time, high-resolution localized forecasting. Conversely, deep learning architectures—such as Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs)—offer rapid inference but frequently suffer from catastrophic extrapolation failures during out-of-sample extreme events and act as opaque systems lacking physical interpretability.

The proposed architecture, a Spatiotemporal, Physics-Constrained, Uncertainty-Quantified Graph Kolmogorov-Arnold Network (ST-PC-UQ-GKAN), bridges this gap. By abandoning the fixed-node activation paradigm of the standard Multi-Layer Perceptron (MLP) in favor of the learnable, spline-parameterized edge functions of the Kolmogorov-Arnold Network (KAN), this framework theoretically achieves superior expressivity and analytical interpretability[1]. The system ingests upstream L1 solar wind time series as a global forcing condition, mapping it onto a spatially distributed graph of ground magnetometer stations. Temporal dynamics are modeled via KAN-gated recurrent units, while spatial dynamics utilize graph message passing. The network is constrained by empirical SW-M-I energy transfer physics via a residual loss, adaptively balanced to avoid gradient pathologies. Outputs include localized magnetic field variations, derived geoelectric fields via magnetotelluric tensors, and uncertainty bounds calibrated via Adaptive Conformal Inference. Ultimately, the trained network undergoes symbolic regression to distill human-readable, generalized governing equations of localized magnetospheric coupling.

Subsystem 1: Kolmogorov-Arnold Network

Foundations

The Kolmogorov-Arnold Representation Theorem

The architectural foundation of this system rests on the Kolmogorov-Arnold representation theorem, formulated in the late 1950s. The theorem mathematically establishes that any multivariate continuous function $f : [0, 1]^n \to \mathbb{R}$ on a bounded domain can be represented as a finite composition of continuous univariate functions and simple addition[3]:

$$f(x_1, \ldots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^{n} \phi_{q,p}(x_p) \right)$$

where $\phi_{q,p} : [0, 1] \to \mathbb{R}$ and $\Phi_q : \mathbb{R} \to \mathbb{R}$ are continuous univariate functions. While standard MLPs are inspired by the Universal Approximation Theorem—utilizing fixed nonlinear activation functions at nodes and learnable linear weights on edges—KANs invert this paradigm. KANs eliminate linear weight matrices entirely; instead, every edge applies a learnable univariate function, and nodes simply sum the incoming signals[1]. It is critical to note that modern KAN architectures are not exact, direct implementations of the theorem. The original theorem restricts the depth to two layers and width to $2n + 1$, which yields highly non-smooth, fractal-like inner functions that cannot be optimized via gradient descent. Modern KANs generalize this representation to arbitrary depths and widths, replacing the purely theoretical construction with smooth, differentiable functions parameterized by data[1].

B-Spline Parameterization versus Alternative Bases

The original KAN implementation parameterizes the edge functions $\phi(x)$ using a residual activation combined with a B-spline[2]:

$$\phi(x) = w \left(\text{SiLU}(x) + \sum_i c_i B_i(x) \right)$$

where w is a learnable scale, $\text{SiLU}(x)$ acts as the base function, and $B_i(x)$ are B-spline basis functions generated recursively via the Cox-de Boor formula defined over a knot vector $T = \{t_0, t_1, \ldots, t_m\}$ [7]:

$$N_{i,0}(x) = \begin{cases} 1 & \text{if } t_i \leq x < t_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

$$N_{i,k}(x) = \frac{x - t_i}{t_{i+k} - t_i} N_{i,k-1}(x) + \frac{t_{i+k+1} - x}{t_{i+k+1} - t_{i+1}} N_{i+1,k-1}(x)$$

While B-splines offer excellent local control—meaning updates to one region of the input space do not cause catastrophic forgetting in distant regions—the sequential dependency of the Cox-de Boor recursion severely hinders GPU parallelization[9]. This introduces substantial parameter overhead $\mathcal{O}(3KW)$ for K basis functions and W connections, making deep B-spline KANs up to 100 times slower than equivalent MLPs[5]. Consequently, alternative bases have been developed to mitigate these computational bottlenecks.

Basis Function	Mathematical Characteristic	Computational Profile	Applicability to Space Weather
B-Spline (pykan)	Piecewise polynomials over adaptive local grids[2].	High overhead due to recursion; strict sequential memory limits[9].	Best for final symbolic extraction, but prohibitive for deep spatiotemporal graphs.
Chebyshev (ChebyKAN)	Orthogonal polynomials with global support[12].	Highly efficient, native GPU parallelization[12].	Excellent for resolving smooth, continuous coupling dynamics.
Gaussian RBF (FastKAN)	Approximates B-splines using Radial Basis Functions[11].	~1.4x speedup over B-splines, 33% memory reduction[11].	Strong candidate for the core graph network due to efficiency and localized scalar retention.
Wavelet (Wav-KAN)	Captures both high and low-frequency components simultaneously[12].	Moderate overhead, robust to varying spectral densities[15].	Highly suitable for capturing impulsive, high-frequency geomagnetic storm sudden commencements.

For this project, efficient-kan (which vectorizes B-spline calculations) or FastKAN (utilizing

Gaussian RBFs) are the recommended open-source PyTorch implementations to ensure feasibility over large spatiotemporal datasets[12].

Regularization, Pruning, and Limitations

Sparsity-inducing regularization is critical to making KANs prunable, a prerequisite for post-hoc symbolic regression. The standard technique involves penalizing the L_1 norm and the entropy of the activation functions, forcing the network to route information through a minimal number of edges[16]. Pruning is executed by dropping edges where the activation magnitude (e.g., non-linear strength, NLS) falls below a predefined threshold (typically 0.05 to 0.1), condensing the network into a sparse, interpretable graph[12].

Despite theoretical advantages, KANs face known limitations at high data scales and dimensionalities. Claims that KANs "break the curse of dimensionality" lack rigorous, universal mathematical proof for practical implementations[5]. Furthermore, KANs are susceptible to rank collapse and grid saturation, where inputs evolve outside the bounded region of the spline grids during training, necessitating dynamic grid extension protocols[2]. Relative to LSTMs or MLPs, a naive B-spline KAN will fail to scale on the massive dimensionalities inherent to global geophysical forecasting unless strictly regularized and mathematically optimized via alternative bases.

Subsystem 2: Temporal Extension (Temporal KAN)

Incorporating Temporal Dependency

Solar wind-magnetosphere coupling is fundamentally a time-history-dependent process. The magnetosphere acts as a non-linear integrator of solar wind energy, accumulating magnetic flux in the magnetotail during the growth phase of a substorm before explosively releasing it during the expansion phase[21]. A static KAN cannot capture this phase transition.

Temporal dependency is incorporated into the architecture by integrating KAN layers into the gating mechanisms of a recurrent architecture, yielding the Temporal Kolmogorov-Arnold Network (TKAN)[23]. Traditional Long Short-Term Memory (LSTM) networks update their cell states (c_t) and hidden states (h_t) using linear weight matrices within the forget, input, and output gates. In a TKAN, these linear transformations are replaced by KAN layers. For instance, the forget gate f_t transitions from $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$ to:

$$f_t = \sigma\left(\mathrm{KAN}_f([h_{t-1}, x_t])\right)$$

This allows the network to learn highly complex, non-linear gating logic explicitly parameterized by splines, preventing compounding errors and catastrophic forgetting across multi-step time series forecasting[23].

Lookback Window and Forecast Horizon

The selection of the temporal windows must be rigorously grounded in the physics of solar wind propagation and magnetospheric dynamics.
The lookback window must account for two distinct physical delays. First, the transit time of solar wind plasma measured by upstream monitors at the L1 Lagrange point (e.g., ACE, DSCOVR) to the Earth's bow shock varies from 30 to 60 minutes, highly dependent on the bulk velocity ($v_{sw} \in [300, 800]$ km/s). Second, the internal loading/unloading dynamics of the magnetosphere (the Dungey cycle) dictate that the growth phase of a substorm can last anywhere from 30 to 120 minutes prior to auroral onset[21]. Consequently, an appropriate lookback window length is **120 to 180 minutes at a 1-minute cadence**. This provides the TKAN with the full historical context of the interplanetary magnetic field (IMF) B_z southward excursion and the ensuing magnetotail flux accumulation.
For the forecast horizon, predicting localized magnetic perturbations beyond 60 minutes introduces severe chaotic divergence due to the impulsive and non-linear nature of magnetotail reconnection and localized field-aligned currents (FACs)[28]. The architecture should target a simultaneous multi-step forecast horizon of $t + 10$, $t + 30$, and $t + 60$ minutes.

Subsystem 3: Spatial/Graph Extension (Graph KAN)

Graph KAN Integration

Geomagnetic disturbances are highly heterogeneous across the Earth's surface, driven by complex three-dimensional current systems such as the auroral electrojets and the substorm current wedge. To model this localized spatial topology, a Graph KAN (GKAN) architecture must be deployed[3]. GKANs embed the spline-based univariate functions within the message-passing framework of Graph Neural Networks (GNNs).
In a standard message-passing GNN, node representations are updated by aggregating transformed features from neighboring nodes. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ represent the graph of magnetometer stations. A KAN-based node update equation takes the form[30]:

$$h_i^{(l+1)} = \mathrm{KAN}_{update}\left(h_i^{(l)}, \mathrm{AGG}_{j \in \mathcal{N}(i)}\left(\mathrm{KAN}_{message}\left(h_i^{(l)}, h_j^{(l)}, e_{ij}\right)\right)\right)$$

This formulation allows the edges of the physical graph to be modeled by the functional edges of the KAN, enabling flexible, non-linear transformations of node features based on their physical connectivity[4].

Graph Construction Logic

The vertices $\mathcal{V}$ are defined by the physical locations of the SuperMAG ground magnetometers. The node features $x_i^{(t)}$ must encapsulate the local magnetic state and position, including the historical magnetic field vector $(\Delta B_n, \Delta B_e, \Delta B_z)$, geographic

coordinates, and the derived Magnetic Local Time (MLT).

The edges $\mathcal{E}$ dictate the information flow. Construction logic should utilize a k-nearest neighbor (k-NN) approach or a distance-thresholded radius graph. The edge weights e_{ij} must be formulated using the great-circle geographic distance and the relative difference in geomagnetic latitude (MLAT). Given the highly zonal nature of auroral electrojets (currents flowing East-West), stations sharing similar MLATs should exhibit stronger edge weights than stations separated longitudinally[32].

A critical architectural decision is whether the graph should be static or dynamic. The Earth continuously rotates beneath the relatively fixed magnetospheric current systems (which are anchored to the Sun-Earth line). Therefore, the Magnetic Local Time (MLT) of any given station is time-varying. Recomputing the full graph adjacency matrix at every 1-minute time step to reflect a dynamic MLT topology is computationally devastating. The established state-of-the-art approach is to construct a **static geographic graph** (where the adjacency matrix is fixed based on unchanging physical distances) but feed the time-varying MLT and solar zenith angle as **dynamic node features**[33]. Furthermore, the upstream solar wind data, which acts as a global driver rather than a localized node, should be injected into the graph updating sequence via a global cross-attention mechanism rather than being forced into the spatial adjacency matrix.

Implementation within PyTorch Geometric requires subclassing the MessagePassing base class. Instead of utilizing linear layers within the message() and update() functions, these operations are instantiated as KANLayer modules[29].

Subsystem 4: Physics-Informed Constraint Layer

Empirical Coupling Functions and the Physics-Residual Loss

Purely data-driven deep learning models are notoriously brittle when confronted with data outside their training distribution. In geophysics, this manifests as catastrophic under-prediction during extreme, "once-in-a-decade" geomagnetic storms[35]. To ensure the GKAN generates physically realistic out-of-sample predictions, the network must be regularized by a Physics-Informed (PIKAN) loss term[6].

Directly evaluating the full 3D magnetohydrodynamic (MHD) partial differential equations (PDEs) within the loss function is mathematically intractable for real-time, global surface forecasting. Instead, established empirical solar wind-magnetosphere coupling functions serve as differentiable constraints. The most robust of these is the Newell Coupling Function ($d\Phi_{MP}/dt$), which represents the rate of magnetic flux opening at the dayside magnetopause[22]:

$$\frac{d\Phi_{MP}}{dt} = v_{sw}^{4/3} B_T^{2/3} \sin^{8/3}(\theta_c/2)$$

where v_{sw} is the solar wind bulk velocity, $B_T = \sqrt{B_y^2 + B_z^2}$ is the transverse interplanetary magnetic field, and θ_c is the IMF clock angle.
The global energy dissipated into the ionosphere—and by extension, the aggregate localized magnetic perturbations—correlates strongly with this coupling function. The physics-informed constraint mandates that the temporal derivative of the globally aggregated spatial predictions (proxied by the predicted SuperMAG SME index) must not wildly violate the energy input provided by the solar wind. The differentiable physics-residual loss is formulated as:

$$\mathcal{L}_{phys} = \frac{1}{T} \sum_{t} \left\| \frac{\partial \text{SME}_{pred}}{\partial t} - \alpha \cdot \text{Newell}(t - \tau) \right\|_2^2$$

where α is a learned scaling parameter aligning the physical units and τ is the inferred propagation delay.

Adaptive Loss Balancing

Training a network with competing loss terms (e.g., standard data-driven MSE loss alongside a physics-residual loss) inevitably triggers gradient pathologies. The differing magnitudes and stiffness of the gradients cause the optimizer to focus entirely on one objective while ignoring the other, stalling convergence[38].
Several adaptive loss-balancing techniques exist to address this in Physics-Informed Neural Networks (PINNs). GradNorm normalizes gradient magnitudes dynamically during backpropagation[40], while SoftAdapt utilizes a softmax function to weight loss components based on their relative rates of change[41].
However, for multi-scale physical systems, the current state-of-the-art is **ReLoBRaLo** (Relative Loss Balancing with Random Lookback). ReLoBRaLo combines principles from SoftAdapt and GradNorm, mitigating stiffness by adopting a temperature-scaled softmax over historical loss ratios combined with an exponentially decaying random lookback mechanism[42]. The adaptive weight $\lambda_i(t)$ for the i-th loss component at epoch t is updated via[44]:

$$\lambda_i(t) = \alpha \frac{\exp\left(\frac{\mathcal{L}_i(t)}{T \cdot \mathcal{L}_i(t_{init})}\right)}{\sum_j \exp\left(\frac{\mathcal{L}_j(t)}{T \cdot \mathcal{L}_j(t_{init})}\right)} + (1 - \alpha)\lambda_i(t-1)$$

By integrating ReLoBRaLo, the PIKAN is forced to respect the Newell energy constraint without sacrificing the high-frequency localized accuracy required for GIC risk forecasting[43].

Subsystem 5: Uncertainty Quantification

Deterministic point forecasts are fundamentally inadequate for power grid operators, who must balance the severe economic costs of preemptive grid configuration against the

catastrophic risk of transformer failure. The forecasting architecture must therefore output rigorous, well-calibrated uncertainty bounds.

Comparison of UQ Methodologies

Methodology	Theoretical Mechanism	Limitations in Spatiotemporal Context	Suitability
Bayesian Neural Networks (BNNs)	Places prior distributions over network parameters, utilizing Variational Inference or MCMC to estimate posteriors.	Computationally ruinous for deep graph networks; extreme difficulty scaling to millions of parameters.	Very Low
Deep Ensembles	Trains N independent models with varying initializations and aggregates the variance of their predictions.	Multiplies training and inference compute costs by N; highly robust but resource-intensive.	Moderate
Monte Carlo (MC) Dropout	Leaves dropout enabled during inference, generating a distribution from multiple stochastic forward passes.	Tends to be poorly calibrated (under-confident or over-confident) in time-series anomaly detection.	Low
Conformal Prediction	A distribution-free framework that wraps around any point predictor to output statistically guaranteed prediction intervals.	Traditional conformal methods fail under time-series distribution shifts (e.g., quiet conditions to storm	High (with adaptation)

		conditions).	

Adaptive Conformal Inference (ACI)

Given the computational complexity of the GKAN and the severe distribution shifts characteristic of space weather data, Adaptive Conformal Inference (ACI) is the optimal UQ strategy. Introduced by Gibbs and Candès (2021), ACI circumvents the assumption of data exchangeability required by standard conformal prediction, making it explicitly suited for time-series data[47].

ACI dynamically updates a non-conformity threshold α_t at each time step based on the empirical coverage of the preceding steps:

$$\alpha_{t+1} = \alpha_t + \gamma(\text{err}_t - \alpha_{target})$$

where err_t is an indicator variable denoting whether the true observation fell outside the predicted interval, α_{target} is the desired miscoverage rate (e.g., 0.1 for a 90% confidence interval), and γ is a step-size parameter. If the prediction interval misses the truth during the onset of a storm, α_t adjusts rapidly, widening the interval to guarantee marginal coverage despite the underlying distribution shift[47].
Evaluating the calibration of these uncertainty estimates requires the construction of reliability diagrams, which plot the predicted confidence intervals against the observed empirical frequencies. The primary mathematical metric for this evaluation is the Continuous Ranked Probability Score (CRPS), which quantifies the distance between the predicted cumulative distribution function (CDF) and the empirical step function of the observation[49].

Subsystem 6: Multi-Task Output Head

The architecture terminates in a multi-task output head designed to simultaneously yield localized magnetic field perturbations, derived geoelectric fields, and categorical GIC risk alerts.

Joint Loss Formulation

The total loss function $\mathcal{L}_{total}$ is a weighted summation of distinct predictive tasks:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{\Delta B} + \lambda_2 \mathcal{L}_{SME} + \lambda_3 \mathcal{L}_{E} + \lambda_4 \mathcal{L}_{risk} + \lambda_5 \mathcal{L}_{phys}$$

1. **Node-Level Magnetic Perturbation ($\mathcal{L}_{\Delta B}$):** Mean Squared Error (MSE) predicting the

temporal derivative of the horizontal magnetic field vectors dB_n/dt and dB_e/dt at each individual SuperMAG node.

2. **Global Index Reconstruction ($\mathcal{L}_{SME}$):** MSE predicting the global SME index. Rather than an independent prediction, the SME index should be derived globally from the upper and lower envelopes of the localized H -component predictions, enforcing internal topological consistency[22].
3. **Physically-Derived Geoelectric Field ($\mathcal{L}_E$):** The ultimate driver of GICs is not the magnetic field, but the geoelectric field $E(t)$. This is derived mathematically in the frequency domain using the magnetotelluric (MT) impedance tensor $Z(\omega)$ [52]:

$$\begin{bmatrix} E_n(\omega) \\ E_e(\omega) \end{bmatrix} = \begin{bmatrix} Z_{nn}(\omega) & Z_{ne}(\omega) \\ Z_{en}(\omega) & Z_{ee}(\omega) \end{bmatrix} \begin{bmatrix} B_n(\omega) \\ B_e(\omega) \end{bmatrix}$$

   The time-series $E(t)$ is recovered via the Inverse Fast Fourier Transform (IFFT). The network compares this derived $E(t)$ to the ground truth (calculated offline prior to training) using an MSE loss.
4. **Ordinal GIC Risk Classification ($\mathcal{L}_{risk}$):** A cross-entropy loss for an ordinal classifier assigning a distinct risk level based on the peak geoelectric field magnitude at each node. Established thresholds identify Level 0: < 1 V/km (safe), Level 1: $1 - 5$ V/km (moderate risk), Level 2: > 5 V/km (severe risk, half-cycle transformer saturation possible)[52].

Multi-Task Weighting and the Physics Residual

The interaction between the task-specific losses (λ_1 through λ_4) and the physics-residual loss (λ_5) must be managed via the previously established ReLoBRaLo framework. Without adaptive weighting, the network will inevitably overfit to the lower-frequency, high-magnitude global index task ($\mathcal{L}_{SME}$), completely failing to predict the localized, high-frequency, extreme dB/dt spikes critical for GIC risk assessment[38]. The multi-task head relies on ReLoBRaLo to ensure gradients flowing backward from the physics constraint regularize, rather than overpower, the localized spatial forecasts.

Data Infrastructure

Upstream Solar Wind and Downstream Magnetometer Data

The upstream forcing data will be sourced from NASA's Coordinated Data Analysis Web (CDAWeb), specifically the OMNI high-resolution dataset. This provides 1-minute cadence time series of the Interplanetary Magnetic Field (IMF) vectors (B_x, B_y, B_z), solar wind bulk velocity (v_{sw}), and proton density (n_p), mathematically propagated from the ACE and DSCOVR spacecraft at the L1 Lagrange point to the Earth's bow shock[55].
The downstream target data is sourced from the SuperMAG initiative via the supermag.jhuapl.edu API (accessible via the Python client)[56]. SuperMAG provides baseline-removed, geographically rotated (North, East, Vertical components) magnetic perturbations from over 300 ground stations at a 1-minute temporal resolution, alongside the derived SME/SML/SMU auroral electrojet indices[22].

Ground Conductivity and Geoelectric Field Modeling

To compute the geoelectric field target data for Subsystem 6, the magnetic field time series must be convolved with local earth conductivity models. The primary source for this data is the U.S. Geological Survey (USGS) regional conductivity maps and the EPRI Fernberg 1D models[53]. 1D models offer depth-only resistivity profiles, representing the Earth as a layered cake, allowing for the calculation of surface impedance via Cagniard-Wait recursive relations[53]. Where available (primarily within the continental United States), 3D empirical electromagnetic transfer functions (EMTFs) from the USArray magnetotelluric survey will be utilized to account for lateral lateral conductivity variations[53].

Storm Candidate List and Preprocessing

The dataset will span Solar Cycles 23, 24, and the ascending phase of 25 (1998–2024). This provides roughly 26 years of 1-minute cadence data across 100+ high-quality stations, yielding approximately 1.3×10^9 node-level observations. Estimated storage requirements range from 50 to 100 GB in NetCDF/HDF5 format.
Data preprocessing will utilize Principal Component Analysis (PCA) and Interquartile Range (IQR) filtering for artifact removal, followed by linear interpolation for minor data gaps (≤ 5 minutes)[62]. Larger gaps will be strictly masked during backpropagation to prevent hallucinated gradients.

Storm Event	Date	Characteristics	Experimental Role
Halloween Storms	Oct-Nov 2003	Extreme X-class flares, severe grid impacts in Sweden/South	Training / Validation.

		Africa[63].	
St. Patrick's Day	Mar 2015	Largest storm of SC24, intense multi-step injection.	Training / Validation.
Gannon Storm	May 10-12, 2024	Extreme G4/G5 event, deep B_z southward excursion, >1000 mV/km geoelectric fields[35].	**Ultimate Extreme Holdout.**

Validation and Experimental Design

The project's validity hinges on a rigorous, highly structured sequence of ablation and benchmarking experiments.

1. **Core Accuracy Benchmark:** The ST-PC-UQ-GKAN will be evaluated against established deep learning baselines: 1D-CNNs, conventional LSTMs, and Spatial-Temporal Graph Convolutional Networks (STGCNs). Evaluation metrics include Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Pearson Correlation for localized dB/dt and E-field predictions.
2. **Extreme-Event Holdout:** Standard MLPs/LSTMs are expected to fail catastrophically on the May 2024 Gannon storm due to the unprecedented magnitude of the solar wind drivers. This experiment tests whether the PIKAN constraints successfully bound the network's extrapolation to physically realistic, albeit extreme, limits[35].
3. **Spatial Generalization:** Train the network on a random 80% subset of the SuperMAG stations and withhold the remaining 20% geographically. Assess the model's capacity to interpolate GIC risk at "unseen" geographical coordinates utilizing the learned spatial message passing.
4. **Symbolic Regression Validation:** Post-training, the KAN edges will be pruned using sparsity thresholds. The remaining active spline functions will be fed into a symbolic regression engine, specifically **PySR** (Python Symbolic Regression)[65]. This experiment aims to mathematically recover human-readable equations detailing the transition from upstream solar wind to localized magnetic perturbation, thereby proving the interpretability claims of the KAN architecture[2].
5. **Physics-Constraint Ablation:** Train identical architectures with and without the ReLoBRaLo-weighted Newell coupling residual loss. Quantify the degradation in topological consistency and energy conservation when the physics constraint is removed[68].
6. **Uncertainty Calibration:** Construct reliability diagrams plotting the Adaptive Conformal

Inference (ACI) confidence intervals against observed empirical frequencies over the solar cycle. Validation is quantified via the Continuous Ranked Probability Score (CRPS)[49].

7. **Multi-Task Ablation:** Compare the GradNorm-balanced multi-task network against independent single-task networks trained purely on the global SME index or purely on local dB/dt, ensuring no destructive interference occurs across task gradients.

Novelty and Prior Art Assessment

A rigorous assessment indicates that this project constitutes a highly novel, graduate-level contribution to the field of space weather forecasting.

Direct Precedent: The use of LSTMs, CNNs, and MLPs to map upstream solar wind to global indices (such as Kp, Dst, and the SuperMAG SME) is deeply established in the primary literature. Similarly, the convolution of magnetic field time series with 1D/3D MT tensors to derive geoelectric fields is standard utility practice for NERC compliance[53].

Extrapolations: Integrating Graph Neural Networks (GNNs) for spatial magnetometer modeling exists on the periphery of recent (2022-2023) literature, though these applications are largely restricted to basic Graph Convolutional Network (GCN) implementations without rigorous physics constraints. Furthermore, Physics-Informed Neural Networks (PINNs) have been widely explored in fluid dynamics and heat transfer[70], but rarely adapted to the highly stochastic, non-differentiable environment of magnetospheric coupling.

Genuinely Novel Contributions: The integration of Kolmogorov-Arnold Networks (KANs) into a spatiotemporal graph setting (GKAN + TKAN) for geophysical forecasting is an entirely unpublished frontier, given that the foundational KAN architecture was only proposed in May 2024[2]. Fusing this structure with ReLoBRaLo physics-informed loss balancing, Adaptive Conformal Inference, and post-hoc symbolic extraction via PySR creates a wholly novel paradigm for interpretable, physics-constrained space weather prediction.

Feasibility and Risk Assessment

Technical Risks and Failure Modes

The project carries significant computational and mathematical risks that must be proactively mitigated.

1. **Computational Overhead (Highest Risk):** The native B-spline parameterization of KANs induces massive parameter overhead[9]. Attempting to embed standard B-spline KAN layers within a message-passing GNN operating over millions of temporal node-steps will inevitably result in GPU Out-of-Memory (OOM) errors and intractable training times.
   - *Mitigation:* The architecture must abandon standard B-splines in favor of FastKAN (Gaussian RBFs) or ChebyKAN (Chebyshev polynomials) to eliminate the recursive Cox-de Boor bottleneck and enable parallelized GPU execution[11].
2. **Gradient Pathologies:** The juxtaposition of classification tasks, high-frequency spatial regression, and low-frequency physics constraints regularly induces severe gradient stiffness, where the network fixates on one loss term and ignores the others[38].
   - *Mitigation:* Strict, verified implementation of the ReLoBRaLo adaptive weighting

algorithm is non-negotiable[42].

3. **Catastrophic Forgetting / Grid Saturation:** Highly volatile storm data may push inputs outside the bounded regions of the spline grids, causing the activations to saturate and collapse the rank of the network[20].
   - *Mitigation:* Implementation of adaptive grid extension routines during backpropagation to organically widen the support of the basis functions[74].

Phased Implementation Roadmap (6-Month Timeline)

- **Phase 1: Data Infrastructure & Baselines (Weeks 1-4).** Construct automated ingestion pipelines for the SuperMAG API and NASA CDAWeb. Process the 1D/3D USGS impedance profiles and compute the baseline geoelectric field dataset. Train standard LSTM and MLP baselines to establish baseline RMSE metrics.
- **Phase 2: KAN Component Isolation (Weeks 5-8).** Implement FastKAN and ChebyKAN frameworks. Test on a simplified 1D time-series regression task (e.g., predicting the global SME index from Solar Wind). Integrate the PySR pipeline to extract the corresponding 1D symbolic equations, proving the interpretability workflow[67].
- **Phase 3: Spatial Graph Development (Weeks 9-12).** Construct the static geographic PyTorch Geometric graph. Implement the dynamic MLT node features. Train a standard STGCN to validate the spatial message-passing logic independent of KAN integration.
- **Phase 4: System Integration (Weeks 13-16).** Replace the GNN and recurrent components with their KAN equivalents, yielding the ST-GKAN. Wrap the output layers in the Adaptive Conformal Inference (ACI) protocol to generate prediction intervals.
- **Phase 5: Physics-Informed Training (Weeks 17-20).** Integrate the Newell coupling residual loss. Code and debug the ReLoBRaLo algorithm and the GradNorm multi-task head. Execute extreme-event holdout testing, specifically targeting the May 2024 Gannon storm.
- **Phase 6: Analysis, Symbolic Extraction, and Reporting (Weeks 21-24).** Execute all ablation studies. Prune the ST-PC-UQ-GKAN and run PySR to extract 3D spatiotemporal governing equations. Construct reliability diagrams and finalize the technical manuscript for AGU/ISEF submission.

Works cited

1. ABFR-KAN: Kolmogorov-Arnold Networks for Functional Brain Analysis - arXiv, https://arxiv.org/html/2601.00416v1
2. KAN: Kolmogorov–Arnold Networks - arXiv, https://arxiv.org/html/2404.19756v1
3. GKAN: Graph Kolmogorov-Arnold Networks - arXiv, https://arxiv.org/html/2406.06470v1
4. KOLMOGOROV–ARNOLD GRAPH NEURAL NETWORKS - OpenReview, https://openreview.net/pdf?id=udfjje2xXb
5. Kolmogorov-Arnold Networks: A Critical Assessment of Claims, Performance, and Practical Viability - arXiv, https://arxiv.org/html/2407.11075v8
6. A Practitioner's Guide to Kolmogorov–Arnold Networks - arXiv, https://arxiv.org/html/2510.25781v5

7. KAN-Tutorial/1_splines.ipynb at main - GitHub, https://github.com/pg2455/KAN-Tutorial/blob/main/1_splines.ipynb
8. PHYSICS-INFORMED DEEP B-SPLINE NETWORKS - Mitsubishi Electric Research Laboratories, https://www.merl.com/publications/docs/TR2026-046.pdf
9. MatrixKAN: Parallelized Kolmogorov-Arnold Network - arXiv, https://arxiv.org/pdf/2502.07176
10. MatrixKAN: Parallelized Kolmogorov-Arnold Network - arXiv, https://arxiv.org/html/2502.07176v2
11. Stable and Efficient Gaussian-Based Kolmogorov–Arnold Networks - MDPI, https://www.mdpi.com/2227-7390/14/3/513
12. Kolmogorov–Arnold Networks for Small Language Models - arXiv, https://arxiv.org/html/2607.15525v1
13. Chebyshev Polynomial-Based Kolmogorov-Arnold Networks: An Efficient Architecture for Nonlinear Function Approximation - arXiv, https://arxiv.org/html/2405.07200v1
14. Solving the cosmological Vlasov–Poisson equations with physics-informed Kolmogorov–Arnold networks | Monthly Notices of the Royal Astronomical Society | Oxford Academic, https://academic.oup.com/mnras/article/545/4/staf2241/8404154
15. GitHub - RamtinMoslemi/KAN-Papers: A complete list of papers on Kolmogorov-Arnold Networks (KANs) extracted from arXiv, https://github.com/RamtinMoslemi/KAN-Papers
16. Hello, KAN! — Kolmogorov Arnold Network documentation - Ziming Liu, https://kindxiaoming.github.io/pykan/intro.html
17. KAN it rediscovery gravity?. Kolmogorov-Arnold Networks (KAN) is a... | by LuxinZ - Medium, https://medium.com/@zhangluxin0317/kan-it-rediscovery-gravity-eba6d83b4b3f
18. Why is the (KAN) Kolmogorov-Arnold Networks so promising - Medium, https://medium.com/@mryasinusif/why-is-the-kan-kolmogorov-arnold-networks-so-promising-8494242a8bdd
19. Kolmogorov-Arnold Networks: A Critical Assessment of Claims, Performance, and Practical Viability - arXiv, https://arxiv.org/pdf/2407.11075
20. AC-PKAN: Attention-Enhanced and Chebyshev Polynomial-Based Physics-Informed Kolmogorov–Arnold Networks - arXiv, https://arxiv.org/html/2505.08687v2
21. Comparative Dynamics of Substorms and Sawtooth Events - DiMarco - 2026 - Journal of Geophysical Research: Space Physics - DOI, https://doi.org/10.1029/2026JA035415?urlappend=%3Futm_source%3Dresearchgate.net%26utm_medium%3Darticle
22. Products - SuperMAG, https://supermag.jhuapl.edu/products/?tab=view
23. Temporal Kolmogorov-Arnold Networks - TKAN - arXiv, https://arxiv.org/pdf/2405.07344
24. [2405.07344] TKAN: Temporal Kolmogorov-Arnold Networks - arXiv, https://arxiv.org/abs/2405.07344
25. Introducing Temporal Kolmogorov-Arnold Networks (TKANs): A Breakthrough in

Multi-step Time Series Forecasting | Mindplex, https://magazine.mindplex.ai/post/introducing-temporal-kolmogorov-arnold-networks-tkans-a-breakthrough-in-multi-step-time-series-forecasting
26. Temporal Kolmogorov-Arnold Networks (T-KAN) for High-Frequency Limit Order Book Forecasting - arXiv, https://arxiv.org/pdf/2601.02310
27. Investigation of inductive electric fields in Earth's magnetotail using Magnetospheric Multiscale (MMS) multipoint measurement - DiVA Portal, https://www.diva-portal.org/smash/get/diva2:1661129/FULLTEXT01.pdf
28. Evaluation of SuperMAG auroral indices as indicators of substorms and auroral power, https://www.researchgate.net/publication/258496921_Evaluation_of_SuperMAG_auroral_indices_as_indicators_of_substorms_and_auroral_power
29. Kolmogorov–Arnold Graph Neural Networks Preprint. Under review. - arXiv, https://arxiv.org/html/2406.18354v1
30. (PDF) Kolmogorov-Arnold Graph Neural Networks - ResearchGate, https://www.researchgate.net/publication/381736955_Kolmogorov-Arnold_Graph_Neural_Networks
31. A Systematic Benchmark of Physics-Informed Neural Network Architectures for the Stiff Poisson–Nernst–Planck System: Adaptive Loss Weighting and Multi-Scale Resolution - arXiv, https://arxiv.org/html/2606.04125v1
32. An indicator of local auroral electrojet peak intensity and latitude inferred from scalar magnetometer measurements made by the Swarm satellites, https://www.swsc-journal.org/articles/swsc/full_html/2025/01/swsc240047/swsc240047.html
33. Global maps of ground magnetometer data - Birkeland Centre for Space Science, https://birkeland.uib.no/wp-content/uploads/2013/02/99-Waters_JGR-NOA.pdf
34. How to Build a Graph-based Kolmogorov Arnold Network | by Ronan Takizawa | GoPenAI, https://blog.gopenai.com/how-to-build-a-graph-based-kolmogorov-arnold-network-d5b37303f452
35. A Canadian perspective on the May 2024 space weather event, https://www.swsc-journal.org/articles/swsc/full_html/2025/01/swsc250051/swsc250051.html
36. Sun-to-Mud observations of the May and October storms of 2024: impacts on Ireland's Space Weather, https://www.swsc-journal.org/10.1051/swsc/2025044
37. A Generative Augmentation and Physics-Informed Network for Interpretable Prediction of Mining-Induced Deformation from InSAR Data - MDPI, https://www.mdpi.com/2072-4292/18/7/987
38. (PDF) Understanding and Mitigating Gradient Flow Pathologies in Physics-Informed Neural Networks - ResearchGate, https://www.researchgate.net/publication/354486143_Understanding_and_Mitigating_Gradient_Flow_Pathologies_in_Physics-Informed_Neural_Networks
39. When and why PINNs fail to train: A Neural Tangent Kernel perspective - ResearchGate, https://www.researchgate.net/publication/355304640_When_and_why_PINNs_fail

_to_train_A_Neural_Tangent_Kernel_perspective
40. Learning Multi-Task Sparse Representation Based on Fisher Information, https://ojs.aaai.org/index.php/AAAI/article/view/29632/31072
41. SoftAdapt: Techniques for Adaptive Loss Weighting of Neural Networks with Multi-Part Loss Functions - Semantic Scholar, https://www.semanticscholar.org/paper/SoftAdapt%3A-Techniques-for-Adaptive-Loss-Weighting-Heydari-Thompson/5d78dd7780a3629ed4c6ca4da9528b3feec8c54d
42. [2110.09813] Multi-Objective Loss Balancing for Physics-Informed Deep Learning - arXiv, https://arxiv.org/abs/2110.09813
43. Adaptively balanced Poisson-constrained physics-informed neural networks for robust displacement integration in background-oriented Schlieren - Optica Publishing Group, https://opg.optica.org/josaa/abstract.cfm?uri=josaa-43-2-333
44. Advanced Schemes and Tools - NVIDIA Docs, https://docs.nvidia.com/deeplearning/modulus/modulus-v2209/user_guide/theory/advanced_schemes.html
45. Time-Domain 3D Electromagnetic Fields Estimation Based on Physics-Informed Deep Learning Framework, https://faculty.sist.shanghaitech.edu.cn/faculty/zhoupq/publications/Time-Domain_3D_Electromagnetic_Fields_Estimation_Based_on_Physics-Informed_Deep_Learning_Framework.pdf
46. Scientific Machine Learning | Physics-Informed Deep Learning and Loss Balancing - Michael Kraus Anton, https://mkrausai.com/project/scientific-machine-learning-physics-informed-deep-learning-and-loss-balancing/
47. Adaptive Conformal Predictions for Time Series - Proceedings of Machine Learning Research, https://proceedings.mlr.press/v162/zaffran22a/zaffran22a.pdf
48. Adaptive Conformal Anomaly Detection with Time Series Foundation Models for Signal Monitoring - arXiv, https://arxiv.org/html/2604.20122v1
49. Help for package verification - CRAN, https://cran.r-project.org/web/packages/verification/refman/verification.html
50. METplus Practical Session Guide (Version 5.0), https://dtcenter.org/book/export/html/1595
51. Forecasting threshold exceedance of atmospheric variables at a specific location - EGUsphere, https://egusphere.copernicus.org/preprints/2026/egusphere-2026-3111/egusphere-2026-3111.pdf
52. An Integrated Tool for Performing Geomagnetic Disturbance Vulnerability Assessment Studies - Thomas Overbye, https://overbye.engr.tamu.edu/wp-content/uploads/sites/146/2025/01/Conference-Integrated-tool-paper_final_Posted.pdf
53. Impact of 1D vs 3D Earth Conductivity Based Electric Fields on Geomagnetically Induced Currents - Thomas Overbye, https://overbye.engr.tamu.edu/wp-content/uploads/sites/146/2018/08/Shetye_1D3D_ISGT_withCopyright_ARCHIVE.pdf

54. A Review of Geomagnetically Induced Current Effects on Electrical Power System: Principles and Theory - SciSpace, https://scispace.com/pdf/a-review-of-geomagnetically-induced-current-effects-on-2fklt9fkow.pdf
55. SuperMAG, https://supermag.jhuapl.edu/
56. Download Data - SuperMAG, https://supermag.jhuapl.edu/mag/
57. Polar Plots - SuperMAG, https://supermag.jhuapl.edu/rBrowse/
58. 1D Earth Resistivity Models for Selected Areas of Continental United States and Alaska, http://jupiter.ethz.ch/~kuvshinov/For_Dima_Alexeev/150116_Relevant_Papers_for_Revision/1D_profiles_for_North_America.pdf
59. theoretical deduction, analytical solutions and numerical modeling in a one-dimensional Earth Magnetotelluric method - SciELO, https://www.scielo.br/j/rbef/a/6RTmkBPPMnSQ5K8XGWN7Ltg/?lang=en
60. Modeling Geomagnetically Induced Currents From Magnetometer Measurements: Spatial Scale Assessed With Reference Measurements - ResearchGate, https://www.researchgate.net/publication/320206231_Modeling_Geomagnetically_Induced_Currents_From_Magnetometer_Measurements_Spatial_Scale_Assessed_With_Reference_Measurements
61. Comparison Between 1-D and 3-D Geoelectric Field Methods to Calculate Geomagnetically Induced Currents: A Case Study - ResearchGate, https://www.researchgate.net/publication/331808345_Comparison_Between_1-D_and_3-D_Geoelectric_Field_Methods_to_Calculate_Geomagnetically_Induced_Currents_A_Case_Study
62. Plasma Density Estimation from Ionograms and Geophysical Parameters with Deep Learning - EGUsphere, https://egusphere.copernicus.org/preprints/2025/egusphere-2025-3070/egusphere-2025-3070.pdf
63. System-Wide Case Study Assessment of Transformer Heating Due to Geomagnetic Disturbances - Thomas Overbye, https://overbye.engr.tamu.edu/wp-content/uploads/sites/146/2020/06/PID6080859.pdf
64. Estimating the Electric Field From Geomagnetically Induced Currents - Thomas Overbye, https://overbye.engr.tamu.edu/wp-content/uploads/sites/146/2024/04/E_Field_State_Estimator_Using_GIC_Measurements_Digest_Footer_copyright_ARCHIVE.pdf
65. pysr • PySR • astroautomata • Skills • Registry • Tessl, https://tessl.io/registry/skills/github/astroautomata/PySR/pysr
66. Discovering equations from data: symbolic regression in dynamical systems - arXiv, https://arxiv.org/html/2508.20257v2
67. OIKAN: A Hybrid AI Framework Combining Symbolic Inference and Deep Learning for Interpretable Information Retrieval Models - MDPI, https://www.mdpi.com/1999-4893/18/10/639
68. Enhancing Physics-Informed Neural Networks with Domain-aware Fourier

Features: Towards Improved Performance and Interpretable Results - arXiv, https://arxiv.org/html/2603.02948v1
69. LRX-PINN: A Layer-Resolving XNet Physics-Informed Neural Network with Integrated Cauchy Activations for Convection-Dominated Problems | Request PDF - ResearchGate, https://www.researchgate.net/publication/408521916_LRX-PINN_A_Layer-Resolving_XNet_Physics-Informed_Neural_Network_with_Integrated_Cauchy_Activations_for_Convection-Dominated_Problems
70. (PDF) Physics-informed Kolmogorov–Arnold networks to model flow in heterogeneous porous media with a mixed pressure-velocity formulation - ResearchGate, https://www.researchgate.net/publication/394095728_Physics-informed_Kolmogorov-Arnold_networks_to_model_flow_in_heterogeneous_porous_media_with_a_mixed_pressure-velocity_formulation
71. Physics-informed neural network with weighted loss and hard constraints for hyperbolic conservation laws - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC12858974/
72. Physics-Informed Neural Networks in Materials Modeling and Design: A Review - Scribd, https://www.scribd.com/document/1014960663/s11831-025-10448-9
73. Kolmogorov–Arnold Networks for Sensor Data Processing: A Comprehensive Survey of Architectures, Applications, and Open Challenges - MDPI, https://www.mdpi.com/1424-8220/26/8/2515
74. kan package — Kolmogorov Arnold Network documentation - Ziming Liu, https://kindxiaoming.github.io/pykan/kan.html